Program: **Diploma in Automation and Robotics**	
Course Code : RAPC 201	Course Title: **Analog & Digital Electronics**
Semester : **3**	Credits: **3**
Course Category: **Engineering Science**	
Periods per week: **3 (L:3, T:0, P:0)**	Periods per semester: **45**

Course Objectives:

- To introduce various semiconductor diodes and applicationcircuits.
- To illustrate the working of differenttransistors.
- To introduce various linear and non linear wave shapingcircuits.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basics of electric circuits, and passive components		Fundamentals of electrical and electronics	2
Basics of Mathematics		Mathematics I & II	1 & 2
Basics ofPhysics		Applied Physics I& II	1 & 2

Course Outcomes

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the working principle of Bipolar Junction Transistor and transistor amplifier.	10	Understanding
CO2	Summarize the working principles of JFET, MOSFET and UJT.	11	Understanding
CO3	Design and implement Combinational and Sequential logic circuits.	11	Applying
CO4	Describe the process of Analog to Digital conversion and Digital to Analog conversion.	11	Applying
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	3		2				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (hours)	Cognitive Level
CO1	**Explain the working principle of Bipolar Junction Transistor and transistor amplifier.**		
M1.01	Describe the physical structure of BJT	2	Understanding
M1.02	Explain the working principle of BJT.	2	Understanding
M1.03	Summarize various transistor configurations and gain relations.	2	Understanding
M1.04	Illustrate the characteristics ofBJT	3	Understanding
M1.05	Explain transistor action and amplification	1	Understanding

Contents:

Bipolar Junction Transistors - Structure - NPN and PNP - symbol - unbiased transistor

Transistor biasing - modes of operation - active, saturation and cut off - operation of NPN transistor - transistor currents

Transistor configurations - current gain α,β and γ - relationship between α and β - effect of temperature in leakagecurrent

Input and output characteristics of common base and common emitter configuration - Cut off, Active and Saturation Regions - comparison of three configurations - importance of CEconfiguration

Transistor action - Transistor as amplifier.

CO2	**Summarize the working principle of JFET, MOSFET and UJT**		
M2.01	Describe the physical structure of JFET, MOSFET and UJT	3	Understanding
M2.02	Explain the working principle of JFET, MOSFET and UJT	3	Understanding
M2.03	Illustrate the characteristics of JFET, MOSFET and UJT	4	Understanding
M2.04	Compare BJT and FET	1	Remembering
	Series Test 1	1	

Contents:

Junction Field Effect Transistors: Structure - symbol - operation of N channel JFET - drain characteristics - transfer characteristics - comparison with BJT.

MOSFET: Structure - symbol - principle of operation (N channel depletion type only), drain characteristics - transfer characteristics.

UJT: structure - symbol - equivalent circuit - intrinsic stand off ratio - operation - VI characteristics. UJT relaxation oscillator

CO3	**Design and implement combinational and sequential circuits.**		
M3.01	Simplification of logic function using K-map	1	Remembering
M3.02	Explain the working of multiplexer, demultiplexer, adder and subtractor.	2	Understanding
M3.03	Develop BCD adder and Carry look head adder	2	Applying
M3.04	Understand the working of different Latches and Flipflops	3	Applying
M3.05	Design shift registers, synchronous and asynchronous counters using flipflops	3	Applying

Contents:

simplification of logic functions using K-map, minimization of logical functions.Don’t care conditions, Multiplexer, De-Multiplexer/Decoders, Adders, Subtractors, BCD arithmetic, carry look ahead adder.

1 bit memory, Clocked SR JK D flipflops,Shift registers, Synchronous and asynchronous counters. Sequence generator.

CO4	**Describe the process of analog to digital and digital analog conversion**		
M4.01	Describe the process of analog to digital conversion.	3	Understanding

M4.02	Explain the working of various digital to analog convertors	4	Understanding
M4.03	Explain the working of various analog to digital convertors	4	Understanding
	Series Test 2	1	

Contents:

Analog to Digital Conversion: Sampling, Quantization, Encoding

Digital to Analog Convertors: Weighted resistor, R-2R ladder

Analog to Digital Convertors: Counter type, Successive approximation, Dual slope, Example of A to D and D to A convertor ICs

Text /Reference:

T/R	BookTitle/Author
T1	R S Sedha - A Text Book of Applied Electronics - S Chand
R1	N N Bhargava, Kulshreshtha and S C Gupta - Basic Electronics and Linear Circuits- TMH
R2	Robert Boylestad - Electronic Devices and Circuits - PHI
R3	Anand Kumar- Fundamentals of Digital Circuits
R4	David A Bell - Electronic Devices and Circuits - PHI

Online resources:

Sl. No	Website Link
1	https://www.electronics-tutorials.ws
2	https://www.elprocus.com
3	http://www.brainkart.com/menu/electronics-engineering/
4	https://www.electrical4u.com
5	https://www. tutorialspoint.com